



CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI - NCR, INDIA

Advance Python Programming

MCA-372

Assignment – 09

BY

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SUBMITTED TO

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SCHOOL OF SCIENCES

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Create an Data Analysis Website using Streamlit , pandas & matplotlib Library :

```
import matplotlib.pyplot as plt
import pandas as pd
import streamlit as st

st.title("📊 Data Analysis")

data = pd.read_csv("./marksheet.csv")

st.subheader("5 Rows of there data set")
st.write(data.head())

st.subheader(" Data Summary")
st.write(data.describe())

columns = data.columns.tolist()
st.write(columns)

x_axis = st.selectbox("Select X-axis", columns, index=0)
y_axis = st.selectbox("Select Y-axis", columns, index=1)
chart = st.radio("Choose Chart Type", ["Line Chart", "Bar Chart", "Histogram",
"Boxplot", "Pie Chart", "Area Chart", "Scatter Plot"], index=0)

fig, ax = plt.subplots(figsize=(8, 5))

if chart == "Line Chart":
    ax.plot(data[x_axis], data[y_axis], marker='o', linestyle='-')
    ax.set_title("Line Chart")

elif chart == "Bar Chart":
    ax.bar(data[x_axis], data[y_axis], color="skyblue")
    ax.set_title("Bar Chart")

elif chart == "Scatter Plot":
    ax.scatter(data[x_axis], data[y_axis], color="red")
    ax.set_title("Scatter Plot")

ax.set_xlabel(x_axis)
ax.set_ylabel(y_axis)

st.pyplot(fig)
```

Output :

To run the Output for Streamlit you need to write an command in the terminal that is **“streamlit run chart.py”**

Deploy ⋮



Data Analysis

5 Rows of there data set

	id	Name	Gender	Age	Section	Science	English	History	Maths
0	1	Bronnie	Female	13	C	21	81	62	49
1	2	Lemmie	Male	15	B	29	41	17	40
2	3	Danya	Female	14	C	12	87	16	96
3	4	Denna	Female	14	B	15	53	82	33
4	5	Jocelin	Male	14	A	43	6	3	21

Data Summary

	id	Age	Science	English	History	Maths
count	15	15	15	15	15	15
mean	8	14.0667	48.6667	48.5333	53.5333	59.2667
std	4.4721	0.7988	32.9841	30.7591	30.7707	24.7602

Deploy

75%	11.5	15	79	77.5	83	78
max	15	15	100	93	96	96

▼ [

```
0 : "id"
1 : "Name"
2 : "Gender"
3 : "Age"
4 : "Section"
5 : "Science"
6 : "English"
7 : "History"
8 : "Maths"
```

]

Select X-axis

id ▼

Select Y-axis

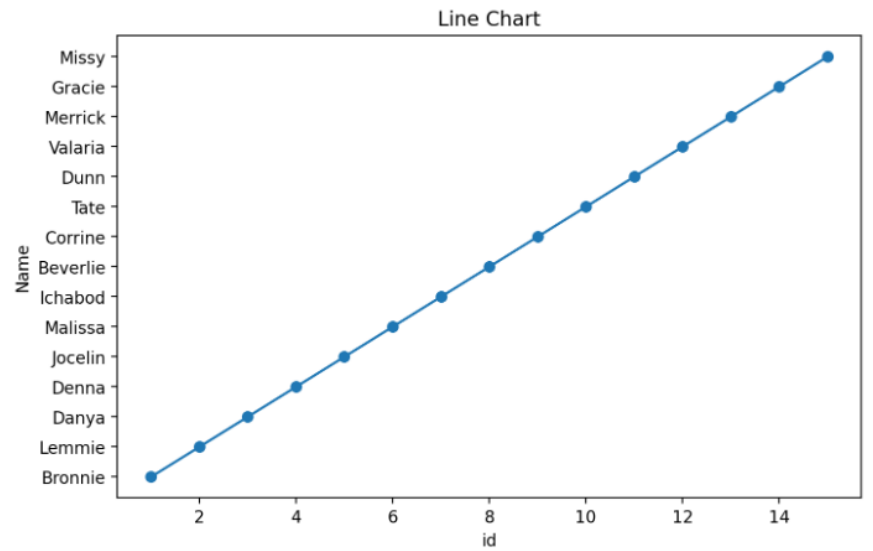
Name ▼

Choose Chart Type

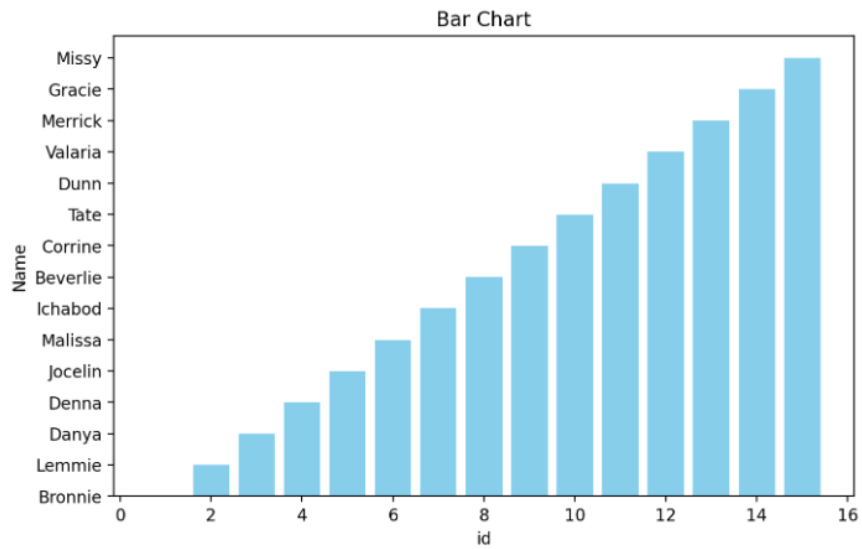
☒ Line Chart

Deploy

- ☐ Bar Chart
- ☐ Histogram
- ☐ Boxplot
- ☐ Pie Chart
- ☐ Area Chart
- ☐ Scatter Plot



- ☒ Bar Chart
- ☐ Histogram
- ☐ Boxplot
- ☐ Pie Chart
- ☐ Area Chart
- ☐ Scatter Plot



- ☒ Scatter Plot

